PREDICTING HEATING AND COOLING LOADS OF BUILDINGS USING MACHINE LEARNING MODELS

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Abstract

Accurate prediction of building heating and cooling loads is important for improving energy efficiency in building design. Machine learning regression models offer an effective approach for estimating thermal loads from building design parameters, although challenges such as feature correlations, multiple related targets, and potential nonlinear relationships may affect predictive performance. This study evaluates several regression models using the Energy Efficiency dataset, including Ordinary Least Squares (OLS), Ridge, Lasso, multi-output linear regression variants, and Support Vector Regression (SVR) with linear and radial basis function (RBF) kernels. Hyperparameters are selected using 5-fold cross-validation, and performance is assessed using MAE and RMSE. Experimental results show that linear regression models produce similar prediction errors, while multi-output linear models achieve comparable performance to their single-output counterparts. Among all evaluated models, the RBF SVR achieves the best performance, with MAE values of 0.3416 for heating load and 0.9128 for cooling load, and RMSE values of 0.4663 and 1.5500, respectively. These findings indicate that nonlinear kernel-based models are more effective in capturing complex relationships between building design parameters and thermal loads.

Keywords: Heating Load, Cooling Load, Machine Learning, Regression, Energy Efficiency, SVR.

1 Introduction

Improving building energy efficiency has become an important objective in modern architectural design and sustainable development. Buildings account for a significant portion of global energy consumption, particularly through heating and cooling systems required to maintain indoor thermal comfort. As a result, accurately estimating the heating load and cooling load of buildings during the design phase is crucial for reducing energy usage and improving overall energy efficiency.

Traditional approaches for evaluating building energy performance often rely on detailed physical simulations or energy modeling software. While these methods can provide accurate estimates, they typically require complex modeling procedures and substantial computational resources. In many practical scenarios, especially during early design stages, fast predictive models that approximate energy performance based on building design parameters can provide valuable guidance for architects and engineers.

Machine learning techniques offer an alternative approach by learning predictive relationships directly from data. In particular, regression-based models can estimate heating and cooling loads

using structural design features such as relative compactness, surface area, wall area, roof area, glazing area, and orientation. However, several challenges arise in this context. Building design parameters often exhibit strong correlations due to geometric dependencies, which may affect the stability of traditional linear regression models. In addition, heating and cooling loads are closely related but not identical targets, raising the question of whether joint multi-output modeling can improve predictive performance. Furthermore, the relationship between building design features and thermal loads may contain nonlinear patterns that are not fully captured by linear models.

To address these challenges, this study investigates the predictive performance of several regression models under different modeling assumptions. Specifically, the analysis compares ordinary least squares (OLS), regularized linear models (Ridge and Lasso), multi-output linear regression variants, and Support Vector Regression (SVR) models with both linear and radial basis function kernels. These models allow systematic examination of three key aspects of predictive modeling: the effect of regularization under correlated predictors, the impact of jointly modeling multiple outputs, and the benefits of nonlinear regression methods.

The main contributions of this study are threefold:

- A systematic comparison of linear, regularized, and kernel-based regression models for predicting heating and cooling loads in building design.
- An evaluation of whether regularization techniques improve model stability when input features exhibit strong correlations.
- An investigation into whether nonlinear models such as kernel-based SVR can significantly improve prediction accuracy compared with traditional linear regression approaches.

2 Related work

2.1 Linear regression

2.1.1 Single-output Linear Regression

Single-output linear regression is one of the most classical and widely adopted regression techniques in the field of energy analytics, particularly for modeling building energy performance due to its simplicity, interpretability, and low computational cost. In this setting, the relationship between a set of building design parameters and a single target variable is assumed to be linear.

Formally, given an input feature vector

$$\mathbf{x} = [x_1, x_2, \ldots, x_d]^\top \tag{1}$$

where $x_1, x_2, \ldots, x_d$ represent building design attributes such as relative compactness, surface area, wall area, roof area, glazing area, and orientation, the single-output linear regression model predicts a scalar response y (e.g., heating load or cooling load) as:

$$y = \mathbf{w}^\top \mathbf{x} + b \tag{2}$$

where $\mathbf{w} \in \mathbb{R}^d$ denotes the vector of regression coefficients and b is the bias term. The parameters are

typically estimated by minimizing the Residual Sum of Squares (RSS):

$$\min_{\mathbf{w},b} \sum_{i=1}^{N} \Big(y_i - (\mathbf{w}^\top \mathbf{x}_i + b) \Big)^2 \tag{3}$$

where N is the number of samples.

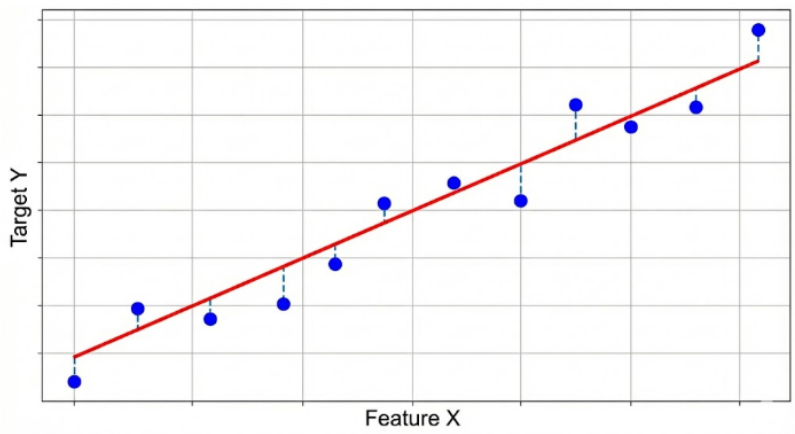

Figure 1: Illustration of single-output linear regression, showing the linear relationship between an input feature and a single target variable.

In building energy prediction, single-output linear regression is commonly used as a baseline model, where separate regressors are trained independently for heating load and cooling load. This formulation provides straightforward interpretation of the influence of individual design parameters on thermal loads.

However, several limitations arise in this context. First, the model assumes weak correlations among input features, whereas building design variables often exhibit strong interdependencies, leading to multicollinearity and unstable coefficient estimates. Second, training independent models for heating and cooling loads ignores potential relationships between the two targets. Finally, the absence of regularization may result in large coefficients and reduced generalization performance when predictors are highly correlated.

These limitations motivate the use of regularized and multi-output regression models, which are discussed in the following sections.

2.1.2 Regularized Linear Regression for Single-Output Prediction

To address the instability of ordinary least squares (OLS), particularly under multicollinearity and potential overfitting, regularization techniques are commonly incorporated into linear regression models. Regularized regression extends the OLS objective by introducing a penalty term that constrains the magnitude of the regression coefficients.

Given a training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{N}$, where $\mathbf{x}_i \in \mathbb{R}^d$ denotes the input feature vector and $y_i \in \mathbb{R}$ represents the target variable, the regularized objective can be written as:

$$\min_{\mathbf{w},b} \frac{1}{N} \sum_{i=1}^{N} (y_i - \mathbf{w}^\top \mathbf{x}_i - b)^2 + \lambda R(\mathbf{w}) \tag{4}$$

where $\lambda > 0$ controls the strength of regularization and $R(\mathbf{w})$ denotes the regularization function. Two commonly used regularization strategies are Ridge and Lasso regression.

Ridge regression applies an ℓ_2-norm penalty:

$$R_{Ridge}(\mathbf{w}) = \|\mathbf{w}\|_2^2 \tag{5}$$

which shrinks coefficient magnitudes toward zero, thereby reducing variance and improving stability when predictors are highly correlated.

Lasso regression employs an ℓ_1-norm penalty:

$$R_{Lasso}(\mathbf{w}) = \|\mathbf{w}\|_1 \tag{6}$$

which encourages sparsity in the coefficient vector by driving some parameters exactly to zero. This property enables implicit feature selection and improves model interpretability.

In building energy prediction tasks, these regularization techniques help mitigate the effects of multicollinearity among design variables while improving the robustness and generalization ability of linear models.

2.1.3 Multi-output linear regression

In many building energy prediction tasks, multiple related targets must be estimated simultaneously. In this study, both heating load and cooling load are predicted from the same set of building design parameters, leading to a multi-output regression formulation.

Let $\mathcal{D} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^N$ denote a dataset where $\mathbf{x}_i \in \mathbb{R}^d$ represents the input features and $\mathbf{y}_i \in \mathbb{R}^m$ denotes the vector of target variables. In this work, $m = 2$, corresponding to heating load and cooling load.

Under the linear modeling assumption, the relationship can be expressed as:

$$\mathbf{y} = \mathbf{W}^\top \mathbf{x} + \mathbf{b} \tag{7}$$

where $\mathbf{W} \in \mathbb{R}^{d \times m}$ is the coefficient matrix and $\mathbf{b} \in \mathbb{R}^m$ is the bias vector. The model parameters are typically estimated by minimizing the aggregated prediction error across all outputs.

However, without additional constraints, the optimization problem often decomposes across output dimensions, effectively learning independent models for each target. To improve robustness under correlated predictors and stabilize parameter estimation, regularization techniques can be incorporated into the multi-output formulation.

The regularized multi-output objective can be written as:

$$\min_{\mathbf{W}} \|\mathbf{Y} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda \Omega(\mathbf{W}) \tag{8}$$

where $\|\cdot\|_F$ denotes the Frobenius norm and $\Omega(\mathbf{W})$ represents a regularization term. Two common regularization strategies are:

Multi-output Ridge regression

$$\Omega(\mathbf{W}) = \|\mathbf{W}\|_F^2 \tag{9}$$

which shrinks coefficients toward zero and improves stability when input features are highly correlated.

Multi-output Lasso regression

$$\Omega(\mathbf{W}) = \|\mathbf{W}\|_1 \tag{10}$$

which promotes sparsity in the coefficient matrix and enables implicit feature selection.

These formulations provide a more stable and robust framework for predicting multiple thermal loads simultaneously.

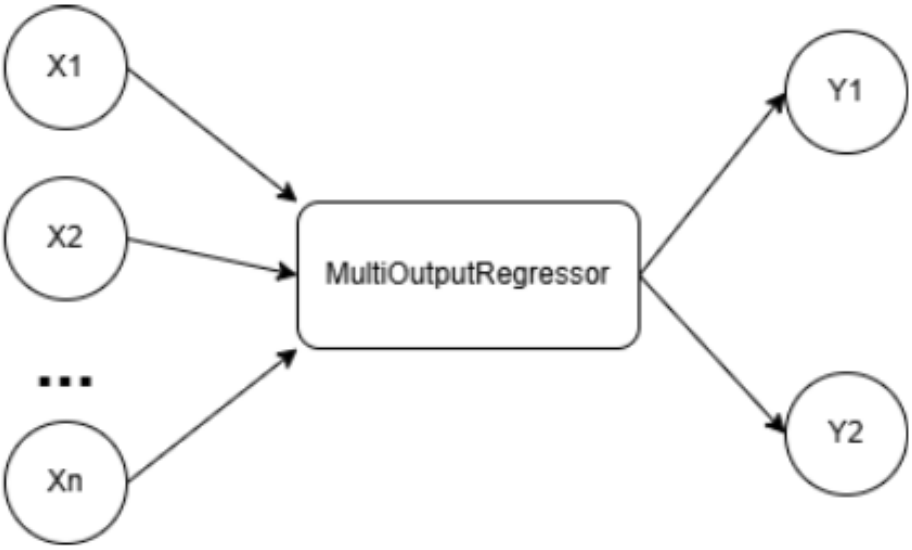

Figure 2: Illustration of the multi-output regression model used in this study, where the input features $X_1, X_2, \ldots, X_n$ are used to predict two target variables Y_1 and Y_2.

2.2 Support Vector Regression

Support Vector Regression (SVR) is a powerful alternative to traditional linear regression models, particularly for datasets exhibiting complex or non-linear relationships. Unlike ordinary least squares (OLS), which minimizes the sum of squared residuals, SVR aims to learn a function that approximates the target values while maintaining model flatness and robustness to noise.

Given a training dataset $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$, SVR seeks to estimate a regression function that deviates from the true targets by no more than a predefined tolerance ε for as many samples as possible. Deviations within the ε-insensitive region are ignored, while larger deviations are penalized using slack variables. This formulation introduces robustness against small fluctuations and measurement noise commonly observed in building energy datasets.

2.2.1 Linear SVR

In the linear case, SVR assumes a linear relationship between the input features and the target variable, expressed as:

$$f(\mathbf{x}) = \langle \mathbf{w}, \mathbf{x} \rangle + b \tag{11}$$

where $\mathbf{w}$ denotes the weight vector, b is the bias term, and $\langle \mathbf{w}, \mathbf{x} \rangle$ represents the inner product between the weight vector and the input feature vector.

The key innovation of SVR lies in the ε-insensitive loss function, which defines a margin (or tube) around the regression function. Prediction errors falling within this tube are not penalized, whereas deviations outside the margin incur a loss proportional to their distance from the boundary. This mechanism improves robustness compared to standard linear regression.

Despite its advantages, linear SVR remains limited by the assumption of linear relationships between predictors and target variables, which may restrict its ability to capture complex interactions among building design parameters.

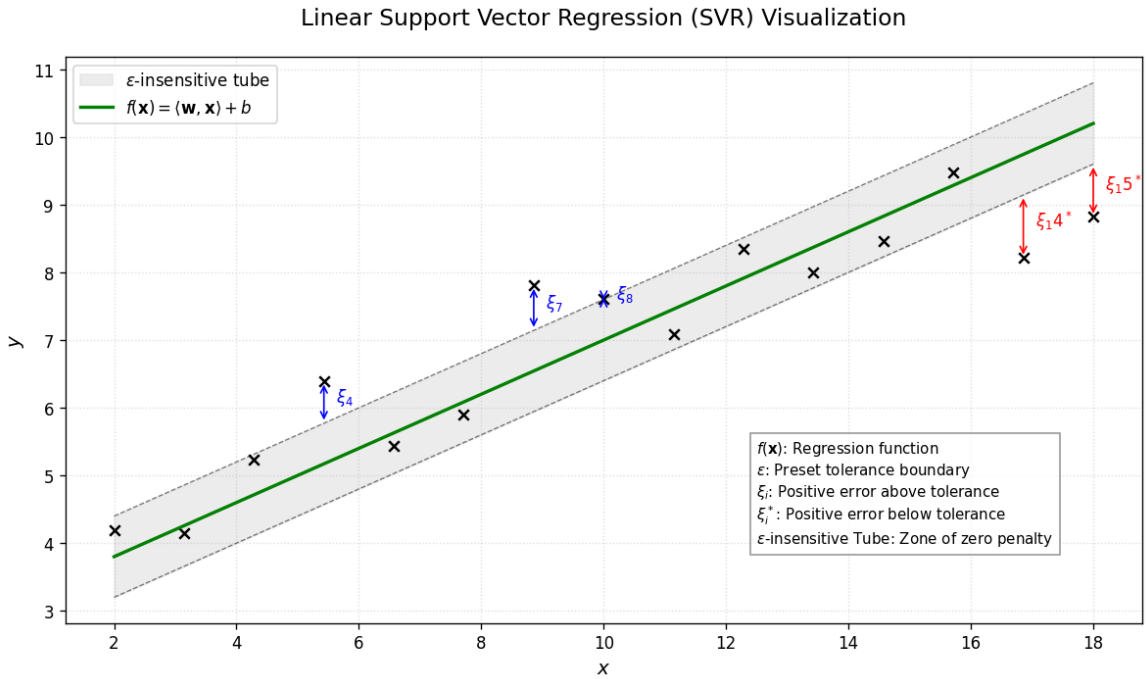

Figure 3: Illustration of linear Support Vector Regression showing the regression function, the ε-insensitive tube, and slack variables.

2.2.2 Non-linear SVR (Kernel-based SVR)

To address non-linear relationships, SVR can be extended through kernel methods that implicitly map the input features into a higher-dimensional feature space. In this transformed space, a linear regression function is learned while representing non-linear relationships in the original input domain.

Among various kernel functions, the Radial Basis Function (RBF) kernel is widely used due to its flexibility and strong empirical performance. The RBF kernel is defined as:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp\left(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|^2\right) \tag{12}$$

where: $\mathbf{x}_i, \mathbf{x}_j$ are the input feature vectors (e.g., building design attributes of samples i and j), $K(\mathbf{x}_i, \mathbf{x}_j)$ is the kernel function, which measures the similarity between two input feature vectors, $\|\mathbf{x}_i - \mathbf{x}_j\|^2$ is the squared Euclidean distance between the two feature vectors in the original space, γ is a hyperparameter that defines how far the influence of a single training example reaches.

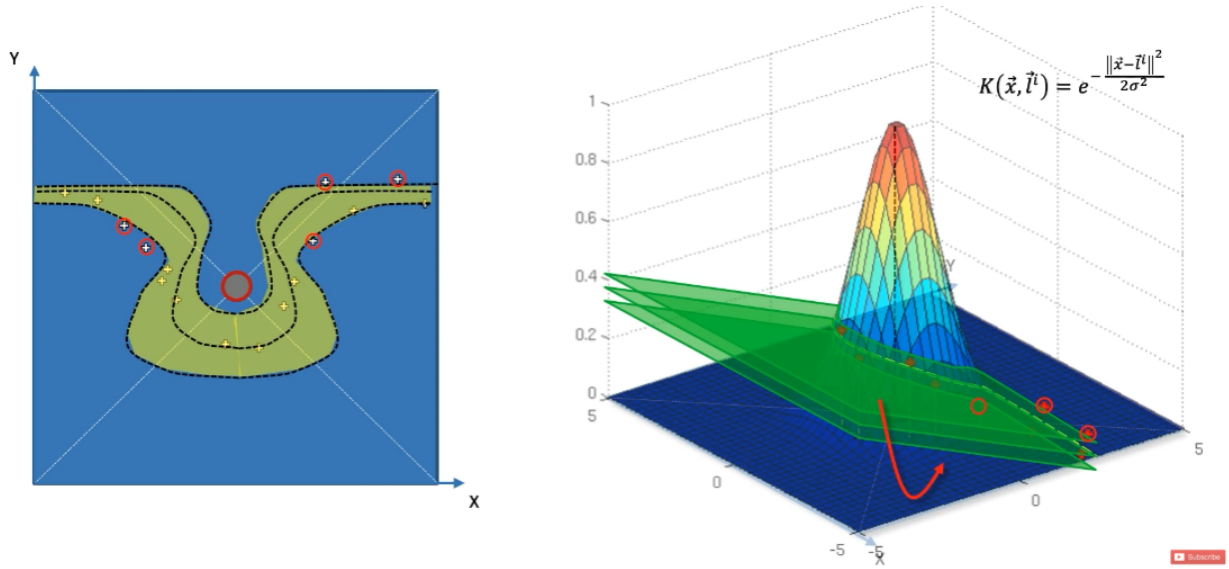

Figure 4: Visualization of non-linear SVR using the kernel trick: (Left) Non-linear regression tube in the original 2D feature space; (Right) Mapping of input data into a high-dimensional feature space using an RBF kernel where a linear hyperplane is constructed.

By capturing non-linear dependencies between building design parameters and thermal loads, kernel-based SVR models are capable of modeling complex energy performance patterns that may not be adequately represented by linear regression approaches.

3 Main contribution

3.1 Problem formulation

In this study, building energy load prediction is formulated as a supervised regression problem with two continuous targets. Let

$$\mathbf{X} \in \mathbb{R}^{n \times p} \tag{13}$$

denote the input feature matrix containing n building configurations described by p design parameters. The corresponding target matrix is

$$\mathbf{Y} \in \mathbb{R}^{n \times 2} \tag{14}$$

where each row $\mathbf{y}_i = (HL_i, CL_i)$ represents the heating load (HL) and cooling load (CL) of the i-th building configuration.

The objective is to learn a mapping from the design feature space to the thermal load variables by minimizing the empirical risk

$$\min_{f \in \mathcal{H}} \frac{1}{n} \sum_{i=1}^{n} \mathcal{L}(\mathbf{y}_i, f(\mathbf{x}_i)) \tag{15}$$

where $\mathcal{L}(\cdot)$ is a regression loss function.

A straightforward formulation treats heating load and cooling load as two independent single-output regression tasks,

$$f_{HL} : \mathbb{R}^p \to \mathbb{R}, \quad f_{CL} : \mathbb{R}^p \to \mathbb{R} \tag{16}$$

However, since both targets originate from the same physical building system and share common design characteristics, they may exhibit statistical dependence. To capture this relationship, an alternative

formulation considers a joint multi-output regression model

$$f : \mathbb{R}^p \to \mathbb{R}^2 \tag{17}$$

where both outputs are predicted simultaneously using a shared parameter structure.

Beyond the output structure, the mapping function may be either linear or non-linear. Linear models assume a global affine relationship between design features and thermal loads, while kernel-based approaches enable non-linear transformations in a higher-dimensional feature space. Accordingly, this study compares models along two dimensions: (1) independent versus joint output modeling, and (2) linear versus non-linear functional forms.

3.2 Data engineering

The dataset consists of 768 simulated building configurations described by eight numerical design parameters and two continuous targets representing heating load (HL) and cooling load (CL). Since all variables are numerical and no missing values are present, preprocessing primarily focuses on preparing the data for stable model estimation.

Feature standardization is applied to all input variables to address differences in scale among predictors. This preprocessing step is particularly important for optimization-based algorithms and kernel methods such as support vector regression, while also ensuring a consistent experimental pipeline across all models.

Structural relationships among predictors are examined to identify potential multicollinearity arising from geometric dependencies in building design parameters. This motivates the evaluation of regularized linear models, including Ridge and Lasso regression.

In addition, the relationship between heating and cooling loads is considered, motivating the comparison between independent single-output models and joint multi-output regression formulations. No significant outliers are detected, and all observations are retained for modeling.

3.3 Unified Machine Learning Pipeline

To ensure a fair comparison among the seven predictive models, all experiments are conducted within a unified experimental framework. The objective of this framework is to isolate the effects of key modeling assumptions—namely regularization, output coupling, and functional form—while eliminating confounding factors related to preprocessing and data partitioning.

The dataset is divided into disjoint training and testing subsets. Model parameters are estimated exclusively using the training data, whereas final performance is evaluated on the held-out test set. No information from the test set is used during preprocessing, model fitting, or hyperparameter selection, thereby preventing information leakage.

Within the training set, hyperparameter selection for regularized linear models and Support Vector Regression is performed using 5-fold cross-validation. The cross-validation procedure is applied only to the training data, ensuring that model complexity is determined independently of the test set.

Feature standardization is applied consistently across all models. The scaling parameters (mean and standard deviation) are estimated from the training data and subsequently applied to the validation and test sets.

For single-output approaches, two independent models are trained separately for heating and cooling load prediction. In contrast, multi-output linear regression models estimate a shared coeffi-

cient matrix within a joint optimization framework. Since Support Vector Regression is inherently single-target, it is applied independently to each output while maintaining identical preprocessing and validation procedures.

By enforcing consistent data partitioning, preprocessing, and hyperparameter selection strategies across all models, the experimental framework ensures that performance differences can be attributed to modeling assumptions rather than implementation artifacts.

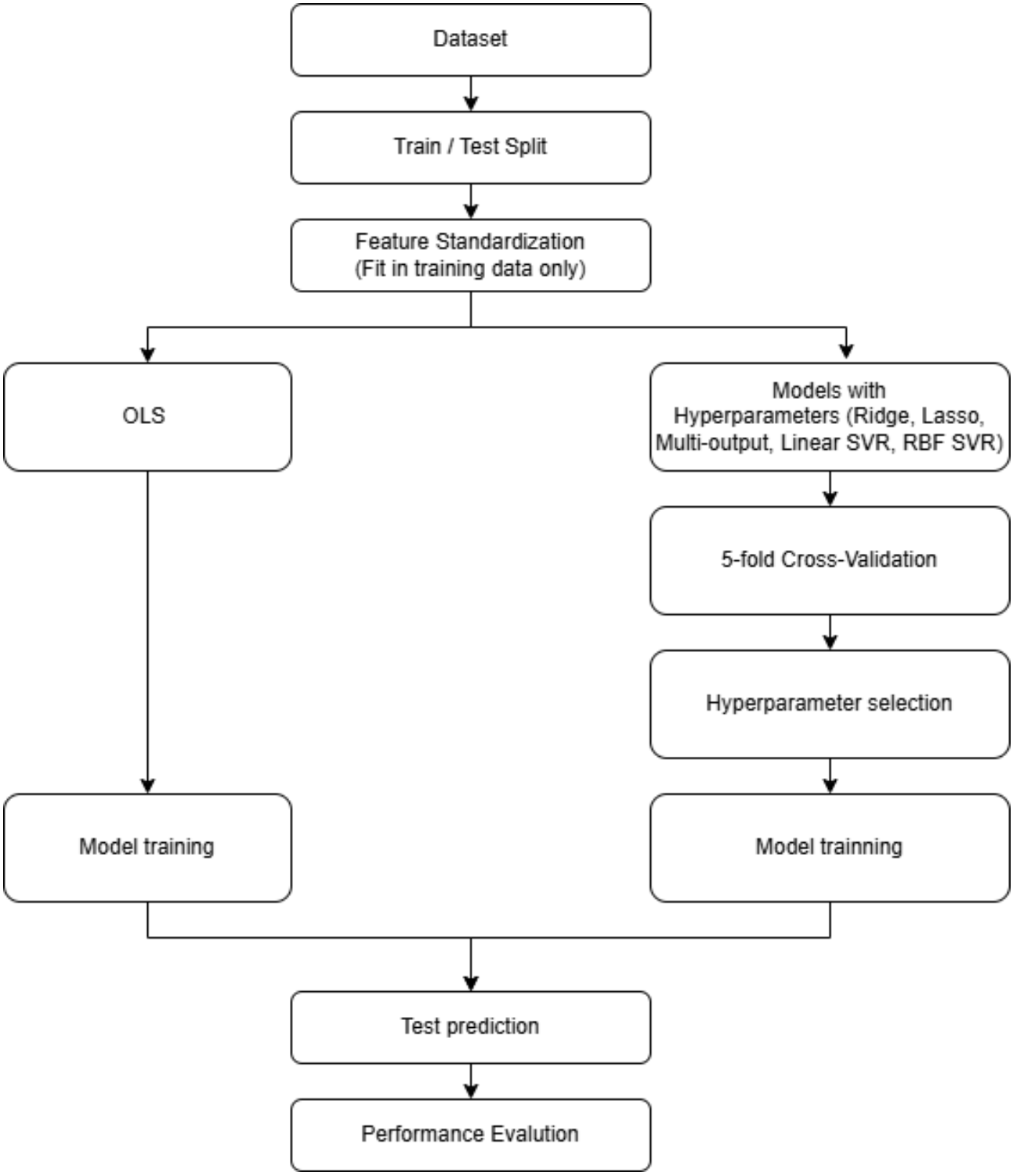

Figure 5: Unified machine learning pipeline used for training and evaluating all predictive models. Hyperparameter tuning via 5-fold cross-validation is applied only to models with tunable parameters, while OLS is trained directly on the training data.

3.4 Model Design and Comparative Framework

To systematically evaluate the effects of regularization, output coupling, and functional form, seven predictive models are implemented within the unified experimental framework. The models are organized into three groups corresponding to distinct structural modeling assumptions. This organization enables controlled comparison while preserving identical preprocessing, data partitioning, and evaluation procedures across all experiments.

3.4.1 Single-Output Linear Models

The first group consists of three single-output linear regression models. In this setting, the standardized feature vector $\mathbf{x} \in \mathbb{R}^p$ serves as input, and the output is a scalar prediction corresponding to either heating load (HL) or cooling load (CL). Separate models are trained independently for each target variable.

Ordinary Least Squares (OLS) serves as the baseline estimator. It determines the regression coefficients by minimizing the residual sum of squares without imposing any regularization constraint. Formally, OLS solves:

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 \tag{18}$$

This model provides a reference point for assessing the impact of regularization and output coupling.

Ridge regression extends OLS by incorporating an ℓ_2-norm penalty on the coefficient vector. The optimization problem becomes:

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_2^2 \tag{19}$$

where λ controls the strength of shrinkage. By constraining coefficient magnitude, Ridge regression mitigates instability caused by multicollinearity among predictors.

Lasso regression further modifies the linear formulation by introducing an ℓ_1-norm penalty:

$$\min_{\mathbf{w}} \|\mathbf{y} - \mathbf{X}\mathbf{w}\|_2^2 + \lambda\|\mathbf{w}\|_1 \tag{20}$$

The sparsity-inducing property of the ℓ_1-norm enables implicit feature selection. Comparing OLS, Ridge, and Lasso allows evaluation of how different regularization strategies influence stability and predictive robustness under independent modeling assumptions.

3.4.2 Multi-Output Linear Models

The second group comprises two multi-output linear regression models, in which heating and cooling loads are predicted simultaneously. In this setting, the input remains $\mathbf{X} \in \mathbb{R}^{n\times p}$, while the output becomes $\mathbf{Y} \in \mathbb{R}^{n\times 2}$. The parameter matrix $\mathbf{W} \in \mathbb{R}^{p\times 2}$ maps input features to both targets.

Multi-output Ridge regression estimates a shared coefficient matrix by solving:

$$\min_{\mathbf{W}} \|\mathbf{Y} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda\|\mathbf{W}\|_F^2 \tag{21}$$

where $\|\cdot\|_F$ denotes the Frobenius norm. The regularization term is applied across the entire coefficient matrix, coupling parameter estimation for both targets. This formulation enables assessment of whether joint learning improves estimation stability compared to training two independent Ridge models.

Multi-output Lasso regression replaces the ℓ_2-penalty with an ℓ_1-based constraint:

$$\min_{\mathbf{W}} \|\mathbf{Y} - \mathbf{X}\mathbf{W}\|_F^2 + \lambda \|\mathbf{W}\|_1 \tag{22}$$

This formulation enforces sparsity across the coefficient matrix while jointly modeling both outputs. Comparing single-output and multi-output variants allows direct evaluation of the impact of output coupling under identical linear assumptions.

3.4.3 Support Vector Regression Models

The final group consists of two Support Vector Regression (SVR) models, which introduce margin-based optimization and the possibility of non-linear functional mappings. Since SVR is inherently a single-target algorithm, separate models are trained independently for heating and cooling load prediction.

Linear SVR assumes a linear relationship between standardized input features and the scalar target variable. The model minimizes an ε-insensitive loss function:

$$\min_{\mathbf{w}} \frac{1}{2}\|\mathbf{w}\|_2^2 + C \sum_{i=1}^{n} \max(0, |y_i - \mathbf{w}^T \mathbf{x}_i| - \varepsilon) \tag{23}$$

where C controls the trade-off between margin width and error tolerance, and ε defines the insensitive region. This formulation differs from OLS and Ridge by emphasizing margin maximization rather than squared error minimization.

Kernel-based SVR with a radial basis function (RBF) kernel extends the linear model by implicitly mapping input features into a higher-dimensional space:

$$K(\mathbf{x}_i, \mathbf{x}_j) = \exp(-\gamma \|\mathbf{x}_i - \mathbf{x}_j\|_2^2) \tag{24}$$

This transformation enables the model to capture non-linear relationships between building design parameters and thermal loads. By comparing linear and kernel-based SVR, the study evaluates whether non-linear modeling provides additional predictive benefit beyond linear structural assumptions.

Collectively, these seven models allow structured comparison along three principal dimensions: the presence or absence of regularization, independent versus joint output modeling, and linear versus non-linear functional assumptions. By maintaining methodological consistency across all experiments, the comparative framework isolates the contribution of each modeling component to predictive performance and estimation stability.

3.5 Evaluation Metrics and Stability Analysis

Predictive performance is evaluated using two standard regression metrics: Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). These metrics are computed independently for heating load (HL) and cooling load (CL) on the held-out test set.

Let y_i denote the true value and $\hat{y}_i$ the predicted value for sample i, with a total of n test observations. The Mean Absolute Error is defined as:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^{n} |y_i - \hat{y}_i| \tag{25}$$

MAE provides a direct measure of the average magnitude of prediction error in the original scale of the response variable. Because it penalizes deviations linearly, MAE reflects typical predictive accuracy without disproportionately weighting extreme errors.

The Root Mean Squared Error is defined as:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{26}$$

By applying a quadratic penalty to deviations, RMSE is more sensitive to large prediction errors. This property allows complementary assessment of model robustness, particularly when evaluating performance across the full range of thermal load values.

The joint use of MAE and RMSE provides balanced evaluation. While MAE captures overall average predictive deviation, RMSE emphasizes error dispersion and sensitivity to substantial misestimation. Reporting both metrics enables comprehensive comparison across modeling assumptions without relying on a single error criterion.

All metrics are reported separately for HL and CL rather than aggregated into a composite score. This design ensures that performance differences across targets are explicitly observable and allows assessment of output-specific sensitivity under independent and joint modeling frameworks.

Hyperparameter selection is conducted exclusively within the training set via cross-validation, and final metrics are computed on an independent test set. Consequently, reported results reflect genuine generalization performance under consistent experimental conditions.

3.6 Training and Testing Strategy

All models are trained using deterministic optimization procedures. Given the moderate dataset size (768 samples), parameter estimation is performed using the full training subset without stochastic or mini-batch updates, ensuring stable convergence and reproducible results.

For Ordinary Least Squares (OLS), model parameters are obtained through analytical least-squares estimation. Ridge and Lasso regressions are trained by solving regularized convex optimization problems. In the multi-output setting, optimization is performed over a shared coefficient matrix that jointly minimizes prediction errors for both heating load and cooling load.

Support Vector Regression (SVR) models are trained using the ε-insensitive loss formulation. Linear SVR operates directly in the standardized feature space, whereas RBF-based SVR employs a kernel function to implicitly map input features into a higher-dimensional space and is optimized in the dual formulation to identify support vectors and corresponding coefficients.

Hyperparameter selection for regularized models and SVR is conducted using 5-fold cross-validation within the training set. For each candidate hyperparameter configuration, the model is trained on four folds and validated on the remaining fold. The configuration yielding the lowest average validation error across folds is selected. After hyperparameter selection, the final model is retrained using the entire training subset.

Final model performance is evaluated on a held-out test set that remains completely unseen during training and hyperparameter tuning. For single-output models—including OLS, Ridge, Lasso, and SVR—two independent models are trained and evaluated separately for heating load and cooling load prediction. In contrast, multi-output linear models produce joint predictions for both targets simultaneously.

Evaluation metrics defined in Section 3.5 (MAE and RMSE) are computed separately for heating load and cooling load using the test set predictions. This evaluation protocol enables consistent comparison of predictive performance across models with different structural assumptions, including regularization, output coupling, and non-linear functional forms.

Random seeds are fixed throughout the experimental process to ensure full reproducibility of the results.

4 Experiment results

4.1 Dataset

The experiments are conducted using the Energy Efficiency dataset, which contains 768 simulated building configurations generated through building energy simulation. Each configuration is characterized by eight continuous design parameters describing geometric and glazing properties of residential buildings.

The input features include relative compactness (X_1), surface area (X_2), wall area (X_3), roof area (X_4), overall height (X_5), orientation (X_6), glazing area (X_7), and glazing area distribution (X_8). These variables represent key architectural factors that influence the thermal performance of buildings.

The prediction targets consist of two continuous variables: heating load (HL) and cooling load (CL), which correspond to the energy required to maintain thermal comfort under different building configurations. Heating load represents the energy demand for indoor heating during cold conditions, while cooling load represents the energy demand for air conditioning during warm conditions.

All variables in the dataset are numerical and no missing values are present. Therefore, no imputation or categorical encoding is required prior to model training. The dataset is randomly divided into training and testing subsets using an 80–20 split to evaluate the generalization performance of the models.

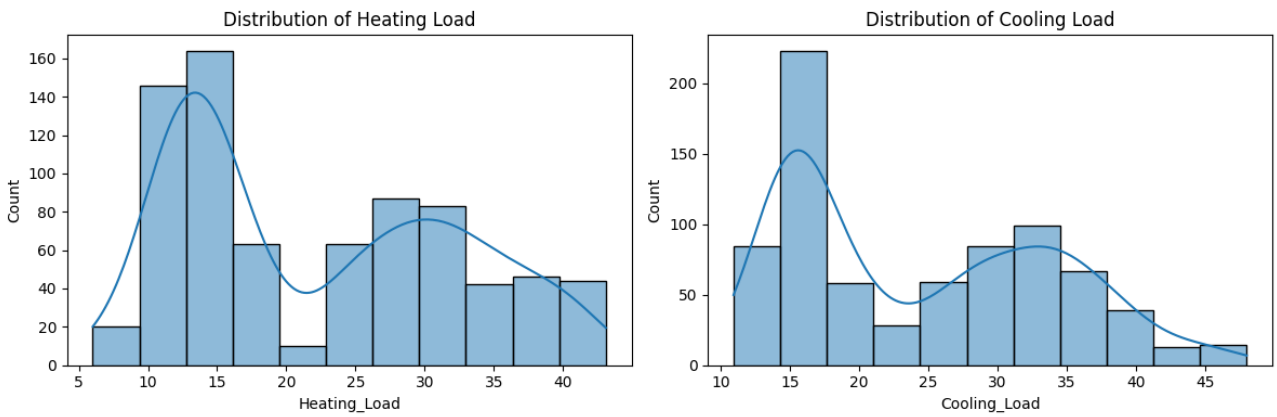

Figure 6: Distribution of heating and cooling loads in the Energy Efficiency dataset.

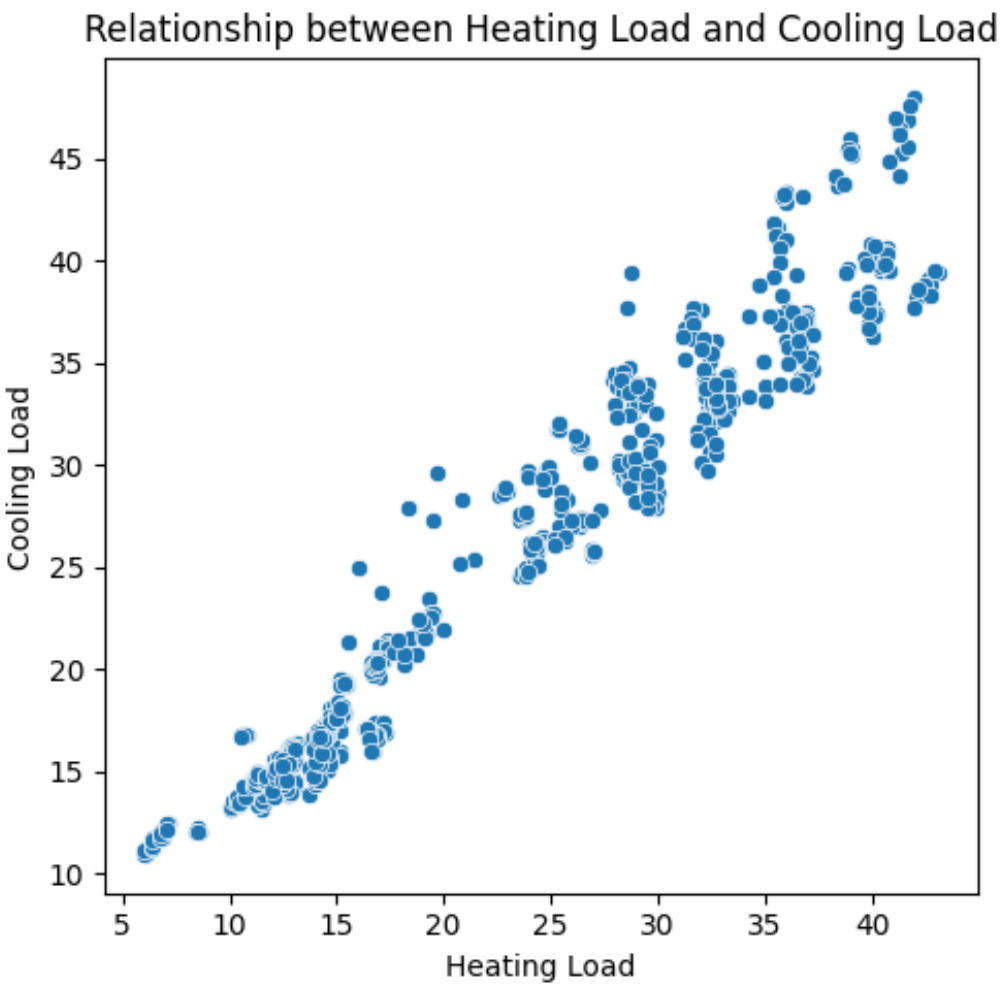

Figure 7: Relationship between Heating Load and Cooling Load.

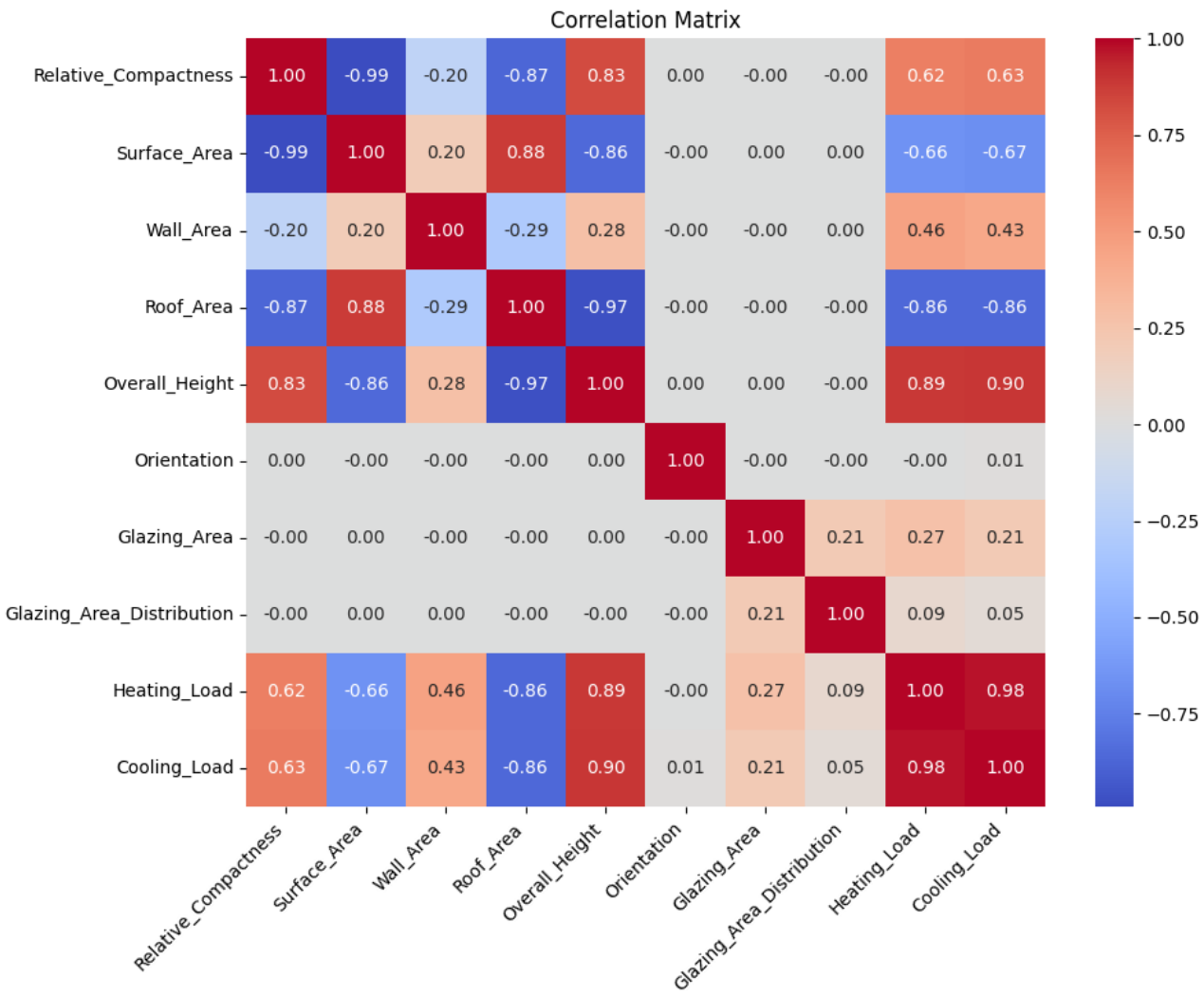

Figure 8: Correlation matrix of input features and target variables.

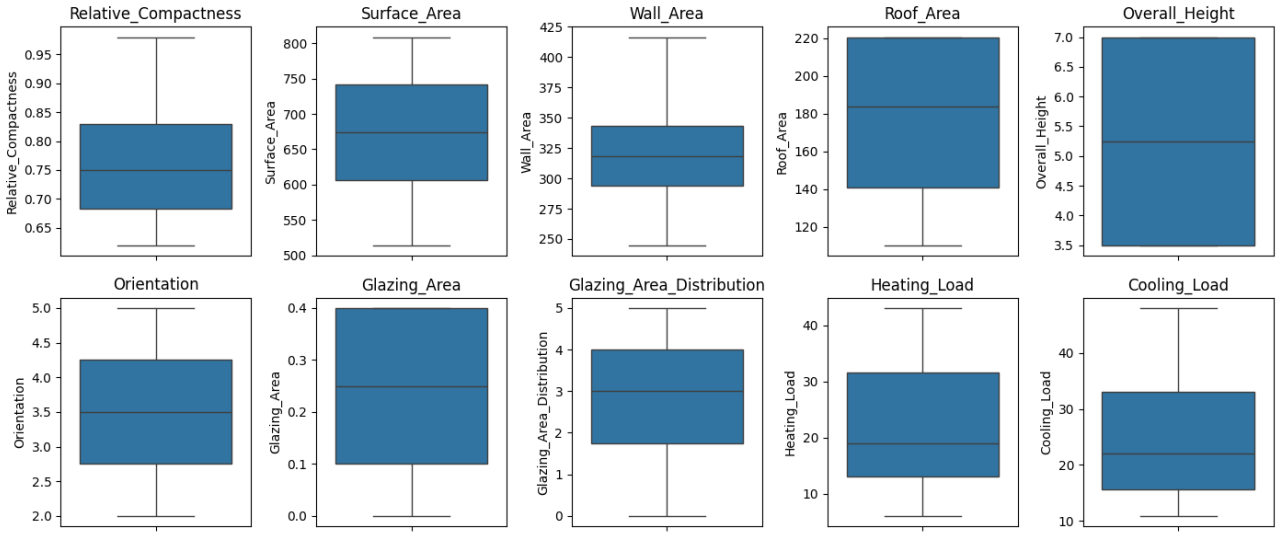

Figure 9: Boxplot distributions of input features and target variables in the Energy Efficiency dataset.

4.2 Results and comparison

The predictive performance of all models is evaluated using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). The results obtained on the test set are summarized in Table 1.

Table 1: Prediction performance of different models.

Model	**HL_MAE**	**HL_RMSE**	**CL_MAE**	**CL_RMSE**
OLS	2.1821	3.0254	2.1953	3.1454
Ridge	2.1829	3.0264	2.1949	3.1457
Lasso	2.1832	3.0262	2.1945	3.1455
Multi-output Ridge	2.1829	3.0264	2.1949	3.1457
Multi-output Lasso	2.1828	3.0259	2.1947	3.1455
Linear SVR	2.1250	3.0679	2.1776	3.2797
RBF SVR	**0.3416**	**0.4663**	**0.9128**	**1.5500**

The results show that linear regression models produce very similar prediction errors. The RMSE values for heating load range from approximately 3.02 to 3.03, while those for cooling load range from 3.14 to 3.15. Linear SVR yields comparable results with RMSE values of 3.0679 for heating load and 3.2797 for cooling load. In contrast, RBF SVR achieves the lowest prediction errors, with RMSE values of 0.4663 for heating load and 1.5500 for cooling load.

Model	**Hyperparameters**
OLS	No hyperparameters
Ridge	$\alpha = 0.1$
Lasso	$\alpha = 0.001$
Multi-output Ridge	$\alpha = 0.1$
Multi-output Lasso	$\alpha = 0.001$
Linear SVR	$C = 10$, $\varepsilon = 1.0$
RBF SVR	$C = 1000$, $\varepsilon = 0.1$, $\gamma = 0.1$

Table 2: Tuned hyperparameters of the evaluated models.

The hyperparameters of each model were optimized using cross-validation on the training set. For Ridge and Lasso models, the regularization parameter α was selected from a predefined grid to control model complexity and prevent overfitting. For SVR models, the penalty parameter C, the ε-insensitive loss parameter ε, and the kernel coefficient γ (for the RBF kernel) were tuned to achieve the best predictive performance. The selected hyperparameters are summarized in Table 2.

5 Discussion

5.1 Linear Modeling and the Effect of Regularization

The first research question investigates whether linear regression models are sufficient for predicting heating load and cooling load. The experimental results show that the three linear regression models—Ordinary Least Squares (OLS), Ridge, and Lasso—produce nearly identical prediction errors.

The RMSE values for heating load range from approximately 3.0254 to 3.0264, while the RMSE values for cooling load range from about 3.1454 to 3.1457. The corresponding MAE values are also very similar across the three models.

These results suggest that introducing regularization does not substantially change the predictive performance of the linear models in this dataset. Ridge and Lasso slightly constrain the magnitude of the regression coefficients, but the overall prediction errors remain very close to those obtained by OLS. This observation indicates that the baseline linear regression model already captures most of the linear relationships between the building design parameters and the thermal load variables.

Overall, the results demonstrate that linear regression models provide reasonably consistent predictions for both heating load and cooling load, serving as a strong baseline for comparison with more complex models.

5.2 Impact of Output Coupling

The second aspect of the analysis examines whether jointly predicting heating load and cooling load improves predictive performance. To evaluate this effect, single-output linear models are compared with their multi-output counterparts.

The results show that the multi-output versions of Ridge and Lasso produce prediction errors that are almost identical to those of the corresponding single-output models. For example, the RMSE of Ridge regression for heating load is 3.0264, which is exactly the same as that obtained by Multi-output Ridge. Similarly, the RMSE values for cooling load remain approximately 3.1457 for both formulations. The differences in MAE are also negligible.

Although heating load and cooling load exhibit a strong correlation in the dataset, the joint learning formulation does not significantly change the overall predictive accuracy compared with training separate models for each target. The numerical results indicate that both modeling strategies yield very similar prediction performance under the linear modeling assumption.

5.3 Impact of Nonlinear Modeling

The final research question investigates whether nonlinear models can improve predictive performance for complex building configurations. This effect is evaluated by comparing linear models, Linear SVR, and kernel-based SVR using the RBF kernel.

The results show that Linear SVR produces prediction errors that are comparable to those of the linear regression models. For heating load prediction, Linear SVR achieves an RMSE of 3.0679, which is close to the values obtained by OLS, Ridge, and Lasso. For cooling load prediction, the RMSE increases slightly to 3.2797.

In contrast, the RBF SVR model achieves substantially lower prediction errors than all other methods. The RMSE decreases to 0.4663 for heating load and 1.5500 for cooling load, representing a significant improvement over both linear regression models and Linear SVR.

These results indicate that nonlinear modeling provides a considerable advantage in capturing the relationships between building design parameters and thermal loads. By implicitly mapping the input features into a higher-dimensional feature space, the RBF kernel allows the model to represent more complex patterns that cannot be captured by purely linear functions.

6 Conclusion

This study investigated the effectiveness of different regression models for predicting building heating load and cooling load using the Energy Efficiency dataset. A unified experimental framework was developed to systematically compare seven predictive models, including OLS, Ridge, Lasso, multi-output linear regression variants, and Support Vector Regression models with both linear and RBF kernels.

The experimental results show that linear regression models, including OLS, Ridge, and Lasso, produce very similar prediction errors, indicating that regularization has only a limited effect on predictive performance in this dataset. The comparison between single-output and multi-output formulations also reveals that jointly modeling heating and cooling loads does not significantly change prediction accuracy. In contrast, nonlinear modeling using RBF-based SVR substantially improves prediction performance, achieving significantly lower MAE and RMSE values for both heating and cooling load prediction.

Overall, the results suggest that while linear models provide a stable and interpretable baseline, nonlinear kernel-based models are more effective in capturing complex relationships between building design parameters and thermal loads.

Future work may explore more advanced machine learning approaches, such as ensemble methods or neural networks, as well as investigate feature engineering techniques that better capture architectural design characteristics influencing building energy consumption.

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